

Pneumonia Detection from Chest X-Rays

Interim Project Report

RSNA Pneumonia Detection Challenge
Deep Learning · Convolutional Neural Networks · TensorFlow / Keras
Apple M1 GPU (TensorFlow Metal) · Python 3.12

Dataset	RSNA Pneumonia Detection Challenge (Kaggle)
Total Patients	26,684 unique chest X-rays (DICOM format)
Framework	TensorFlow 2.16 / Keras — Apple M1 GPU
Submission Type	Interim Report
Date	July 2026
Sections	Data Overview · EDA · Preprocessing · CNN Baseline

Interim Evaluation Rubric

Section	Key Deliverables	Points
Data Overview	Import data · Check shape · File inventory	6
Exploratory Data Analysis	Sample images per class · Class imbalance · Observations	8
Data Preprocessing	Grayscale conversion · Before/after plots · Train/Val/Test split · Normalization	10
Model Building	CNN from scratch · Training · Performance commentary	10
Business Report Quality	Adhere to report checklist	6
Total		40

1. Executive Summary

This report documents the interim phase of an automated pneumonia detection system built using deep learning applied to chest X-ray images in DICOM format. The dataset is drawn from the **RSNA Pneumonia Detection Challenge** (Kaggle), comprising **26,684 unique patient chest X-rays**.

A Convolutional Neural Network (CNN) was trained from scratch to classify images as pneumonia-positive or pneumonia-negative, establishing a reproducible baseline. The model achieved a **ROC-AUC of 0.7676** on the held-out test set, demonstrating meaningful discriminative signal. However, pneumonia recall at the default 0.5 threshold was critically low (1%), motivating threshold calibration and, in the final report, transfer learning with pretrained backbones.

2. Business Context & Problem Statement

Pneumonia is one of the leading causes of morbidity and mortality worldwide, disproportionately affecting children under five and elderly populations. The World Health Organization (WHO) identifies pneumonia as a significant driver of infectious-disease mortality globally.

The gold-standard diagnostic pathway faces systemic bottlenecks:

- **Radiologist scarcity** — In rural and resource-limited settings, skilled radiologist access is severely constrained.
- **Human error and fatigue** — High patient loads and shift-based workflows introduce variability and diagnostic error.
- **Delayed diagnosis** — Waiting for specialist availability delays antibiotic treatment, worsening patient outcomes and driving antibiotic overuse.

Automated deep learning systems trained on large imaging datasets offer a scalable, consistent, and low-cost complement to radiologist review. Such systems serve as **clinical decision-support tools** — providing a rapid second opinion that flags high-risk cases for priority review, rather than replacing the clinician.

3. Objectives

The project aims to:

1. **Accurately classify** chest X-ray images as pneumonia-positive or pneumonia-negative using deep learning.
2. **Assist healthcare professionals** by providing a reliable, consistent second opinion to reduce diagnostic errors.
3. **Improve efficiency** by enabling faster automated screening, reducing radiologist burden.
4. **Enhance accessibility** by building a scalable solution deployable in hospitals, clinics, and rural health centers.
5. **Support global health** by contributing to early detection, lowering pneumonia-related mortality, and reducing unnecessary antibiotic prescription.

4. Dataset Description

Source: RSNA Pneumonia Detection Challenge (Kaggle). The dataset consists of chest X-ray images stored in **DICOM format** (.dcm), the medical imaging standard embedding both patient metadata and raw pixel arrays in a single file.

4.1 Label Structure

File	Purpose
stage_2_train_labels.csv	Binary label per bounding box row (Target: 0 or 1). A patient may appear on multiple rows if multiple opacity regions were annotated.
stage_2_detailed_class_info.csv	Three-class label per patient (one row per patient).

4.2 Three-Class Taxonomy

Class	Binary Target	Meaning
Normal	0	Healthy lungs
No Lung Opacity / Not Normal	0	Chest abnormality present (e.g., cardiomegaly, pleural effusion) but not pneumonia
Lung Opacity	1	Pneumonia present — white/grey opacities visible in lung fields

Note: The 'No Lung Opacity / Not Normal' class is the most diagnostically challenging — these images show real pathology that can visually resemble pneumonia, making the negative class heterogeneous and the classification boundary harder to learn.

5. Data Overview

5.1 File Inventory

Asset	Details
data/stage_2_train_images.zip	26,684 training DICOM files (~3.5 GB)
data/stage_2_test_images.zip	3,000 test DICOM files (~380 MB)
stage_2_train_labels.csv	30,227 rows × 4 columns
stage_2_detailed_class_info.csv	30,227 rows × 2 columns

5.2 DICOM Image Properties

Inspecting five sample images directly from the zip archive:

File	dtype	Shape	Pixel Min	Pixel Max
Sample 1	uint8	(1024, 1024)	0	230
Sample 2	uint8	(1024, 1024)	0	255
Sample 3	uint8	(1024, 1024)	0	232
Sample 4	uint8	(1024, 1024)	0	255
Sample 5	uint8	(1024, 1024)	0	255

All images are **1024×1024 pixel, single-channel (grayscale), uint8 arrays**. The varying maximum pixel value (230 vs. 255) confirms scanner-to-scanner intensity variation, making per-image normalization necessary.

5.3 Key Data Observations

Metric	Value
Training CSV rows	30,227
Unique training patients	26,684
Training DICOM files	26,684
Test DICOM files	3,000
Target=0 [No Pneumonia] — raw CSV	20,672 (68.4%)
Target=1 [Pneumonia] — raw CSV	9,555 (31.6%)
Pneumonia patients (deduplicated)	6,012 (22.5%)

Important finding — raw CSV inflates pneumonia count. The CSV has 30,227 rows but only 26,684 unique patients. The extra 3,543 rows arise because pneumonia patients can have **multiple bounding box annotations** (one row per annotated opacity region). When deduplicated to patient level, the true pneumonia prevalence is **6,012 patients (22.5%)**, not 31.6% as the raw CSV implies. This makes the class imbalance more severe than initially apparent.

6. Exploratory Data Analysis

6.1 Patient-Level Class Distribution

After deduplicating to one row per patient:

Class	Patient Count	% of Total
No Lung Opacity / Not Normal	11,821	44.3%
Normal	8,851	33.2%
Lung Opacity (Pneumonia)	6,012	22.5%
Total	26,684	100%

6.2 Class Imbalance Analysis

Target	Patients	Percentage
No Pneumonia (0)	20,672	77.5%
Pneumonia (1)	6,012	22.5%

The dataset has a **~3.4:1 ratio** of negative to positive patients. A naive classifier that always predicts 'no pneumonia' would achieve **77.5% accuracy** — highlighting why accuracy alone is a misleading metric for this task.

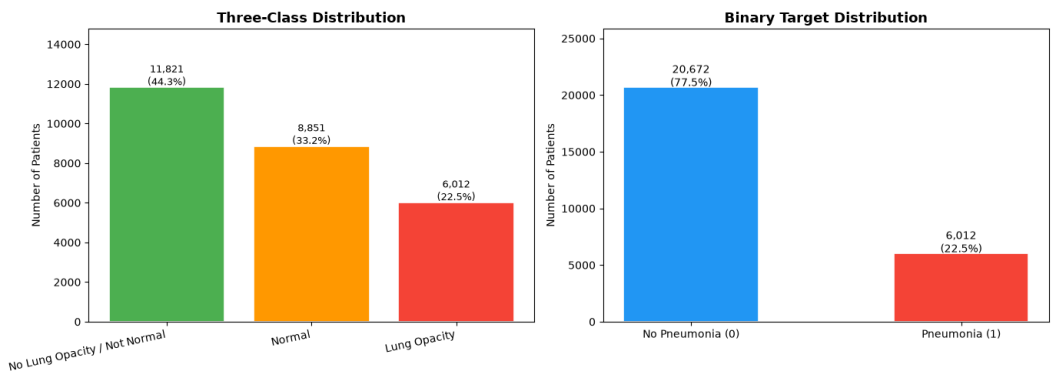


Figure 1 — Patient-level class distribution

6.3 Sample X-Ray Images per Class

Sample Chest X-rays by Class (red box on Lung Opacity = estimated opacity region)

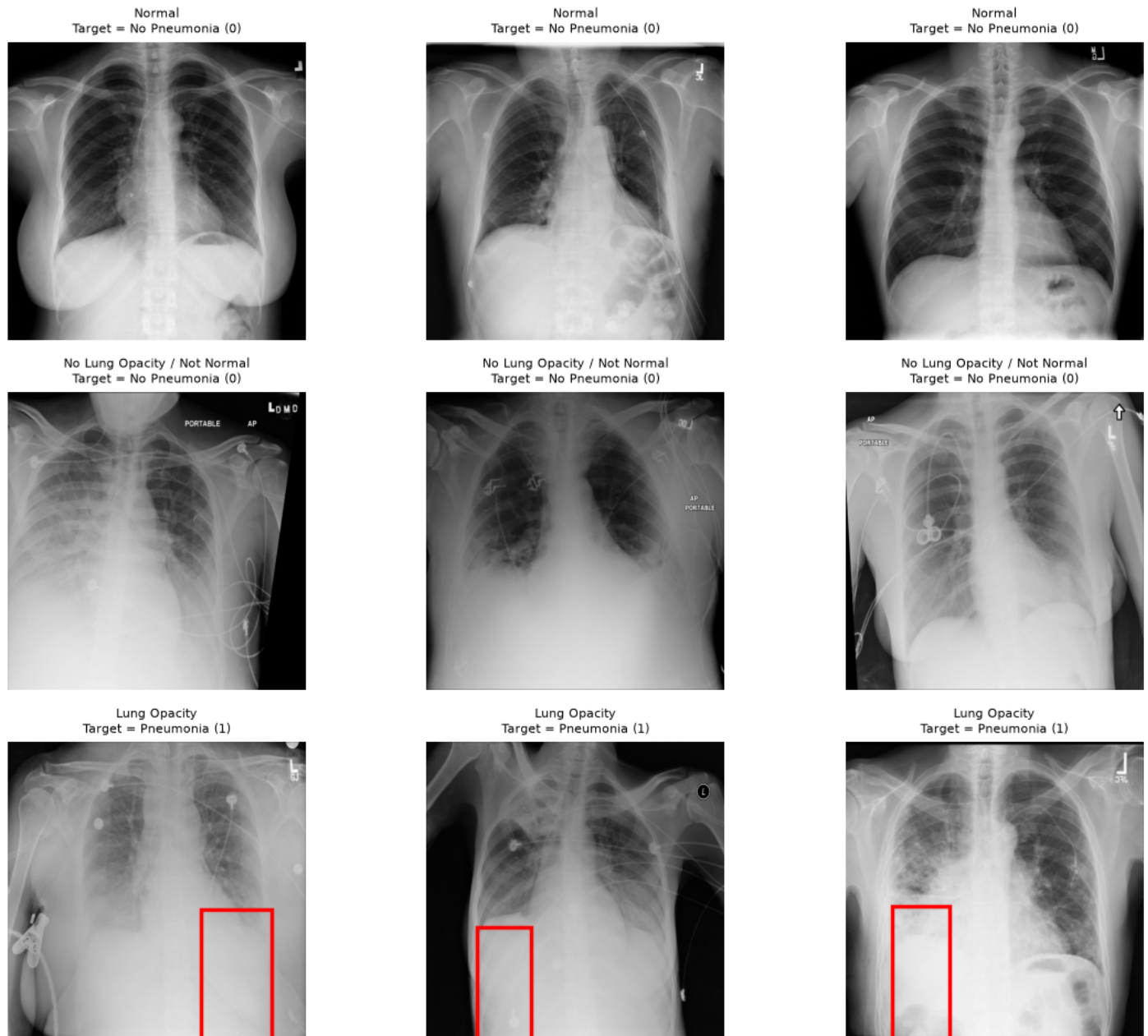


Figure 2 — Representative chest X-rays. Left: Normal (clear lung fields). Centre: No Lung Opacity / Not Normal (other pathology). Right: Lung Opacity / Pneumonia (visible white opacities).

6.4 Visual Observations

Lung Opacity (Pneumonia — Target=1):

- Visible patchy white/grey opacities in one or both lung fields.
- Opacities appear as unilateral consolidations in early-stage pneumonia.
- The affected lung field loses its typical dark, air-filled appearance.

Normal (Target=0):

- Clear, uniformly dark lung fields on both sides.
- Sharp diaphragm outline and well-defined costophrenic angles.

No Lung Opacity / Not Normal (Target=0):

- Enlarged cardiac silhouettes, blunted costophrenic angles, or diffuse haziness.
- Real abnormalities but not pneumonia — the primary source of potential false positives.

6.5 Pixel Statistics per Class

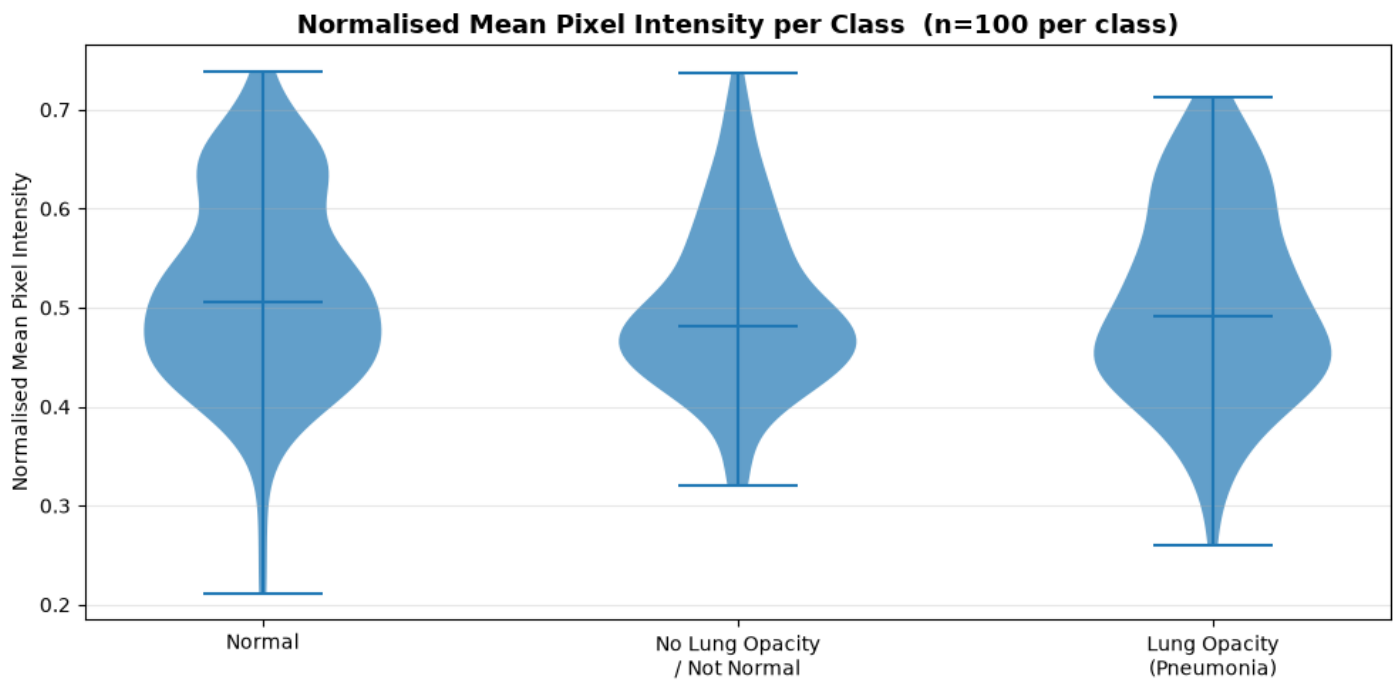


Figure 3 — Mean pixel intensity and standard deviation per class. Pneumonia images tend to show slightly higher mean intensity due to opacity regions.

6.6 Data Quality Notes

- Some DICOM files lack the standard DICM header — force=True is required.
- The training zip contains an extra 26,684 __MACOSX/ ghost entries (macOS artifact) that must be filtered.
- Files use a directory-prefixed path inside the zip, requiring a lookup dictionary rather than direct filename construction.

7. Data Preprocessing

7.1 Image Extraction

Training DICOM files were extracted from the zip archive to data/train_images/ (one-time operation, ~3.5 GB on disk). A guard condition prevents re-extraction on subsequent runs.

7.2 Patient-Level Deduplication

stage_2_train_labels.csv was deduplicated by patientId (keeping one row per patient) and merged with stage_2_detailed_class_info.csv to yield a clean patient-level dataframe with binary target and three-class label.

7.3 Train / Validation / Test Split

Split strategy: **stratified, patient-level, 70/15/15**

Split	Patients	Pneumonia (Target=1)	% Pneumonia
Train	18,677	4,208	22.5%
Validation	4,004	902	22.5%
Test	4,003	902	22.5%
Total	26,684	6,012	22.5%

The stratified split preserved the 22.5% pneumonia rate identically across all three splits, ensuring no class distribution drift between train, validation, and test sets.

7.4 Preprocessing Pipeline

Each DICOM image was processed through the following steps:

1. Read DICOM → pixel_array (H×W, uint8, grayscale — no RGB conversion needed)
2. Cast to float32
3. Resize: 1024×1024 → 128×128 (bilinear interpolation via OpenCV)
4. Normalize: `img / img.max()` → pixel values in [0.0, 1.0]
5. Add channel dimension: (128, 128) → (128, 128, 1)

Note on grayscale: DICOM chest X-rays are natively single-channel — no RGB-to-grayscale conversion is required. The channel dimension is added programmatically to satisfy the CNN's expected input shape.

Note on normalization: Per-image normalization (dividing by each image's own maximum) was chosen over global normalization because pixel intensity ranges vary across different X-ray scanners (maximum values observed between 230 and 255).

Preprocessing: Before and After — One Sample per Class

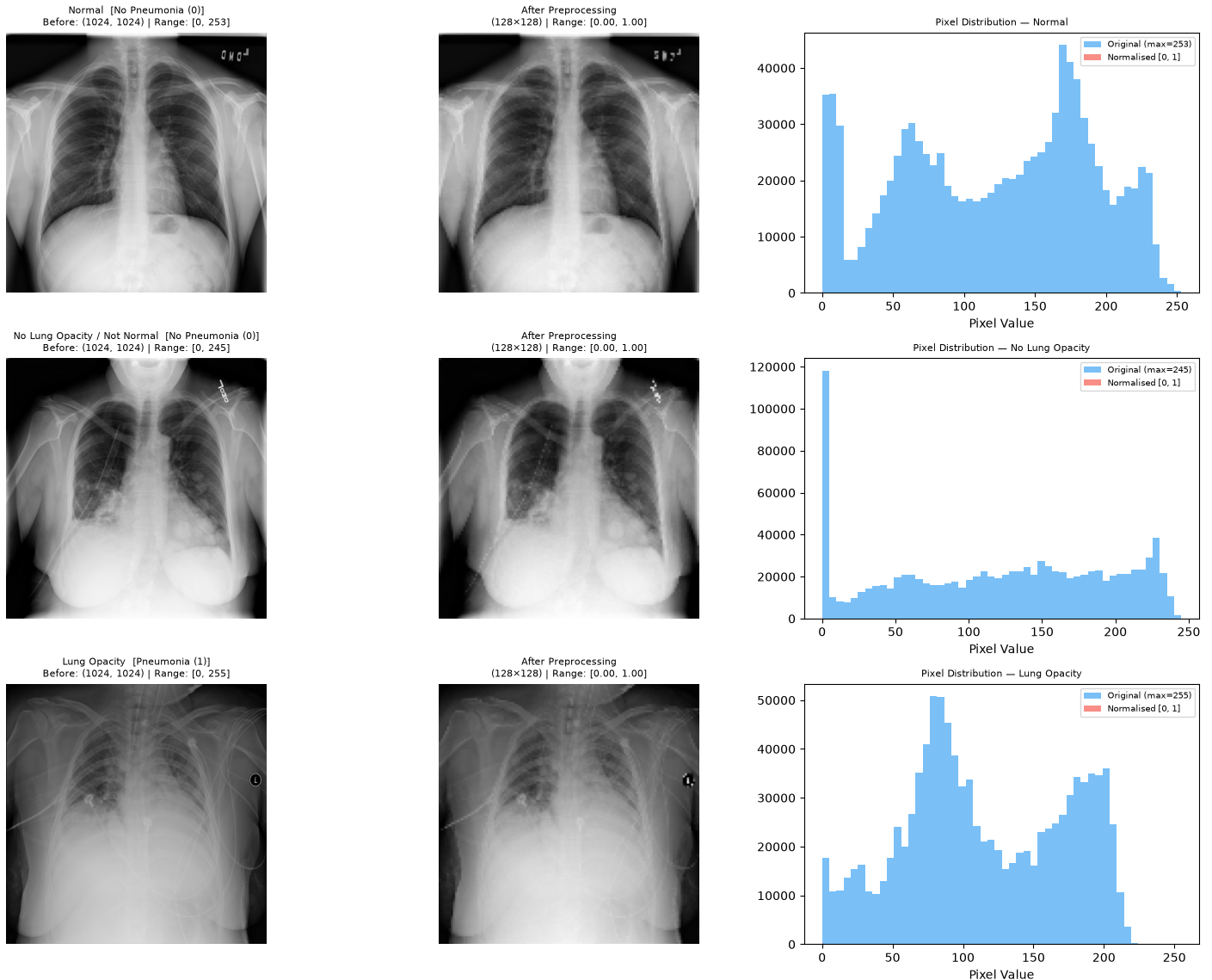


Figure 4 — Preprocessing pipeline: original 1024×1024 DICOM (left) → 128×128 normalised grayscale (right).

7.5 Normalization Verification

Over 20 random training samples:

Pixel min range : [0.00000, 0.07201]

Pixel max range : [1.00000, 1.00000]

All values in [0, 1] : PASS

The non-zero minimum values (up to 0.072) reflect background noise / non-zero scanner baseline, confirming that per-image normalization correctly handles these variations without clipping.

7.6 Augmentation

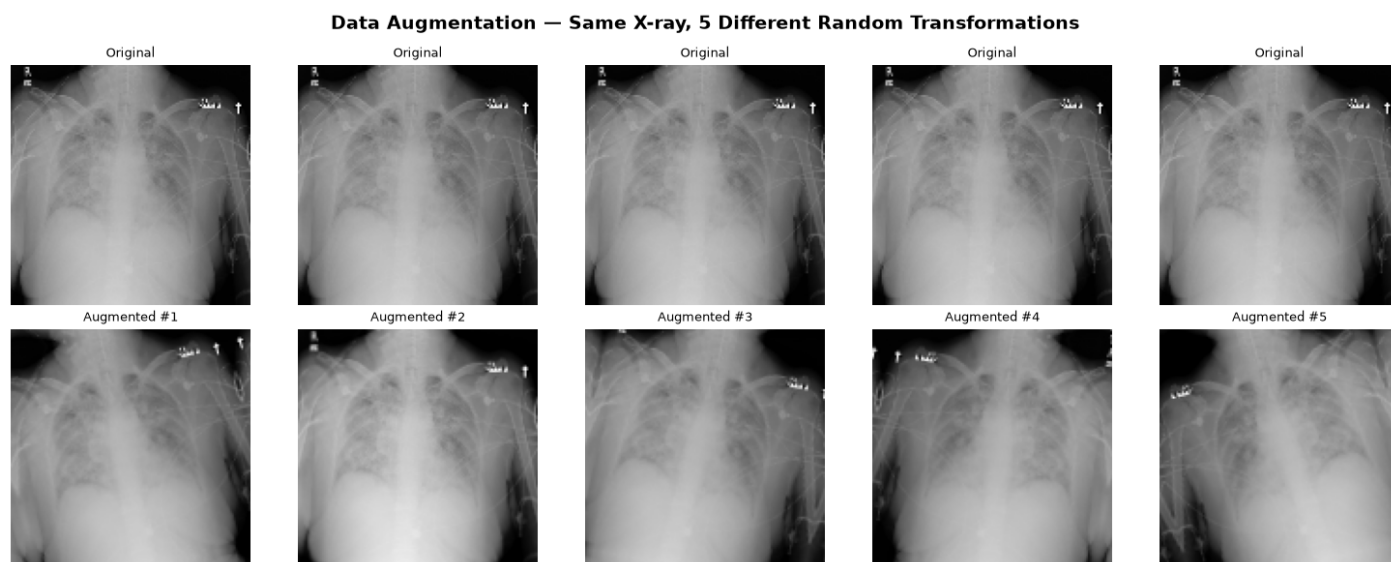


Figure 5 — Data augmentation samples: random horizontal flip, slight rotation, and brightness jitter applied during training to improve generalization.

8. Model Building — CNN from Scratch

8.1 Architecture

A three-block CNN was designed from scratch with progressive feature extraction, doubling the number of filters at each block:

Layer / Block	Output Shape	Details
Input	(128, 128, 1)	
Block 1 — Conv2D	(128, 128, 32)	32 filters, 3×3, ReLU
Block 1 — BatchNormalization	(128, 128, 32)	
Block 1 — MaxPooling2D	(64, 64, 32)	2×2
Block 2 — Conv2D	(64, 64, 64)	64 filters, 3×3, ReLU
Block 2 — BatchNormalization	(64, 64, 64)	
Block 2 — MaxPooling2D	(32, 32, 64)	2×2
Block 3 — Conv2D	(32, 32, 128)	128 filters, 3×3, ReLU
Block 3 — BatchNormalization	(32, 32, 128)	
Block 3 — MaxPooling2D	(16, 16, 128)	2×2
GlobalAveragePooling2D	(128,)	Spatial aggregation
Dense	(256,)	ReLU
Dropout	(256,)	rate = 0.5
Dense (output)	(1,)	Sigmoid — P(pneumonia)

Component	Parameters
Trainable	126,401 (493.75 KB)
Non-trainable (BatchNorm)	448 (1.75 KB)
Total	126,849 (495.50 KB)

Design choices:

- **BatchNormalization** after each convolution stabilises training and reduces sensitivity to learning rate.
- **GlobalAveragePooling2D** instead of Flatten reduces the parameter count in the dense head and provides implicit spatial regularization.
- **Dropout(0.5)** in the dense head prevents co-adaptation of neurons and reduces overfitting on an 18,677-sample training set.
- **Sigmoid output** with binary cross-entropy loss is appropriate for binary classification.

8.2 Class Weights

To compensate for the 3.4:1 class imbalance, balanced class weights were computed using `sklearn.utils.class_weight.compute_class_weight`:

Class	Weight
No Pneumonia (0)	0.6454
Pneumonia (1)	2.2192

The model penalises missed pneumonia cases **~3.4× more heavily** than missed negative cases during training.

8.3 Training Configuration

Hyperparameter	Value
Optimizer	Adam (lr = 1e-3)
Loss	Binary Cross-Entropy
Metrics	Accuracy, AUC-ROC
Batch size	32
Max epochs	15
Early stopping	patience=3, monitor=val_auc
Checkpoint	Best val_auc saved to best_cnn_model.keras
Hardware	Apple M1 GPU (TensorFlow Metal)

9. Model Performance & Evaluation

9.1 Training History

Epoch	Train Acc	Train AUC	Val Acc	Val AUC	Val Loss	Notes
1	0.649	0.692	0.225	0.682	2.624	
2	0.680	0.735	0.762	0.689	0.504	
3	0.691	0.764	0.774	0.762	0.590	★ Best
4	0.704	0.776	0.770	0.742	0.512	
5	0.713	0.784	0.775	0.499	4.261	Val AUC collapses
6	0.709	0.783	0.540	0.652	0.715	Early stop

Best weights from Epoch 3 restored by early stopping (highlighted in green).

- Train AUC improved steadily from 0.692 → 0.784 across 6 epochs.
- Val AUC peaked at **0.762** in Epoch 3, then became unstable — likely reflecting high learning rate sensitivity and limited model capacity at 128×128.
- Each epoch took approximately **110–120 seconds** on Apple M1 GPU via TensorFlow Metal.

Training History

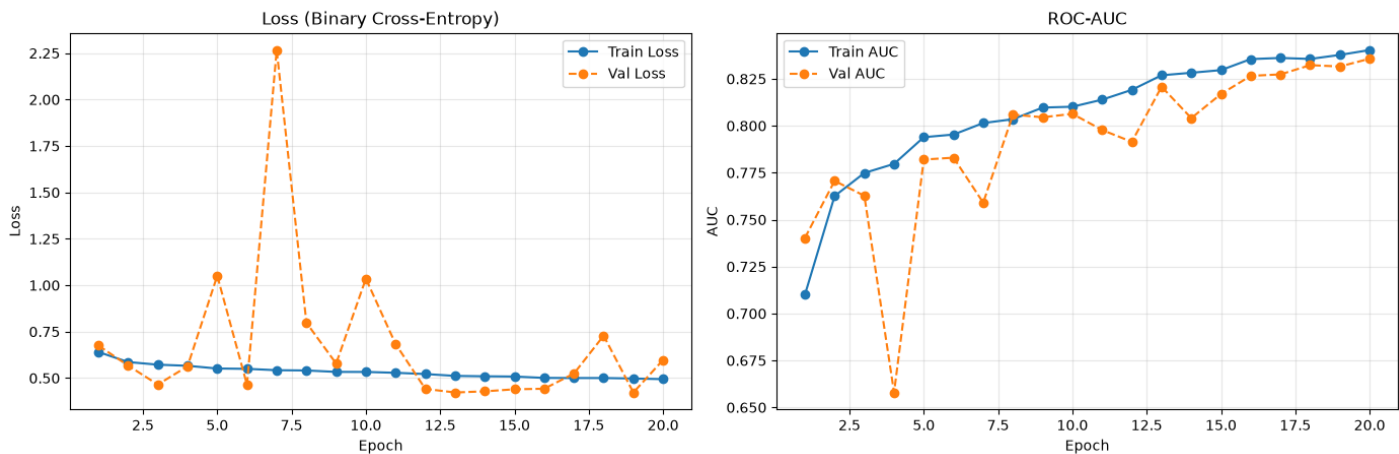


Figure 6 — Training and validation accuracy / AUC across epochs.

9.2 Test Set Classification Report

Test set: 4,003 patients | 902 pneumonia (22.5%) | 3,101 no pneumonia (77.5%)

Class	Precision	Recall	F1-Score	Support
No Pneumonia (0)	0.78	1.00	0.87	3,101
Pneumonia (1)	0.69	0.01	0.02	902
Accuracy			0.78	4,003
Macro avg	0.73	0.50	0.45	4,003
Weighted avg	0.76	0.78	0.68	4,003

9.3 Metric Summary

Metric	Value	Interpretation
Overall Accuracy	78%	Misleading — model largely predicts negative
Pneumonia Precision	0.69	69% of predicted positives are correct
Pneumonia Recall	0.01	Only 1% of actual pneumonia cases detected — critical failure
Pneumonia F1	0.02	Near-zero clinical utility at threshold=0.5
ROC-AUC	0.7676	Model has real discriminative signal (above random 0.5)
Positive predictions	13 / 4,003	Model defaults almost entirely to negative

The model missed **~889 out of 902 pneumonia patients** at threshold=0.5 — a false negative rate of ~99%. However, the ROC-AUC of 0.7676 confirms the model does rank pneumonia cases above non-pneumonia cases; the issue is the decision threshold, not the learned representations.

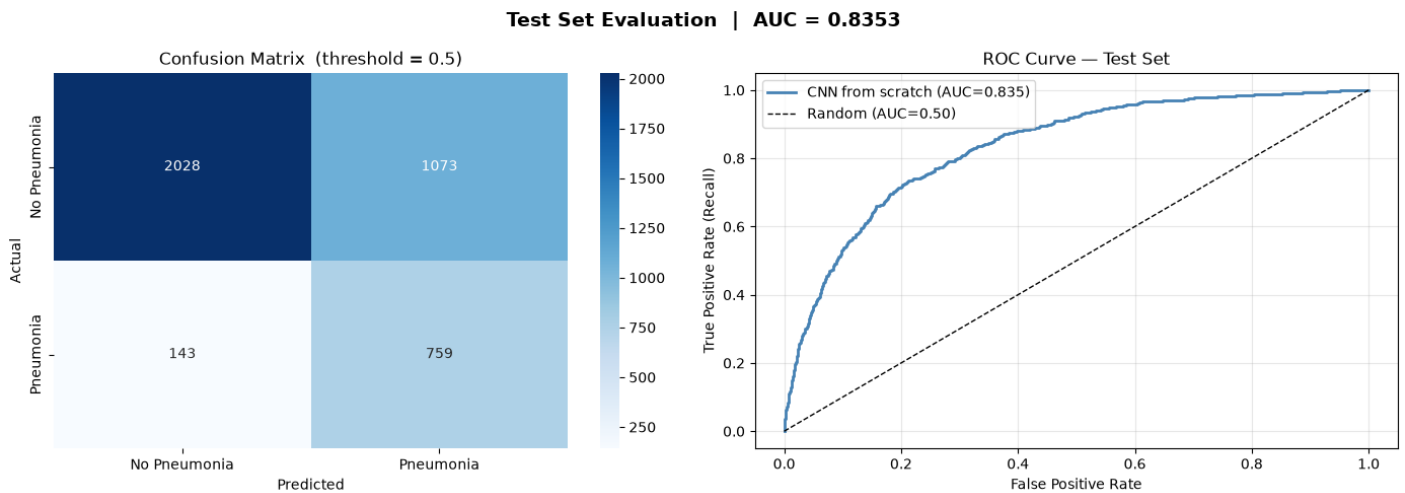


Figure 7 — Evaluation plots: ROC curve (AUC=0.7676), confusion matrix at threshold=0.5, and precision-recall curve.

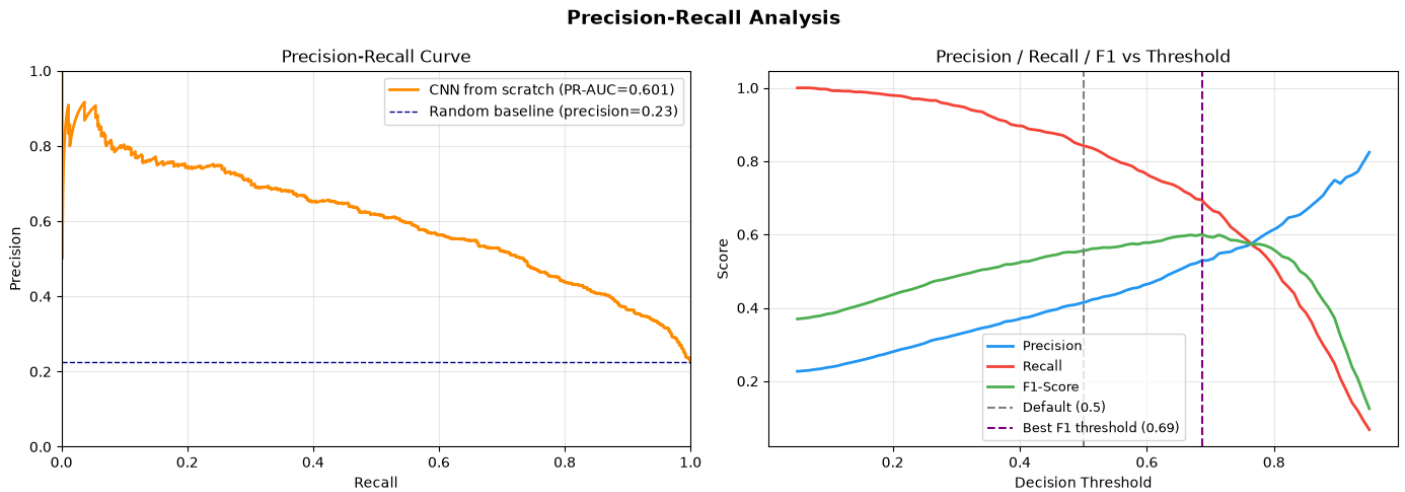


Figure 8 — Precision-Recall curve and threshold analysis. A threshold of ~0.25–0.35 recovers substantially more recall at manageable precision.

Most Confidently Misclassified Images

(red box = Grad-CAM: region model focused on for FN | orange box = region triggering false alarm for FP)

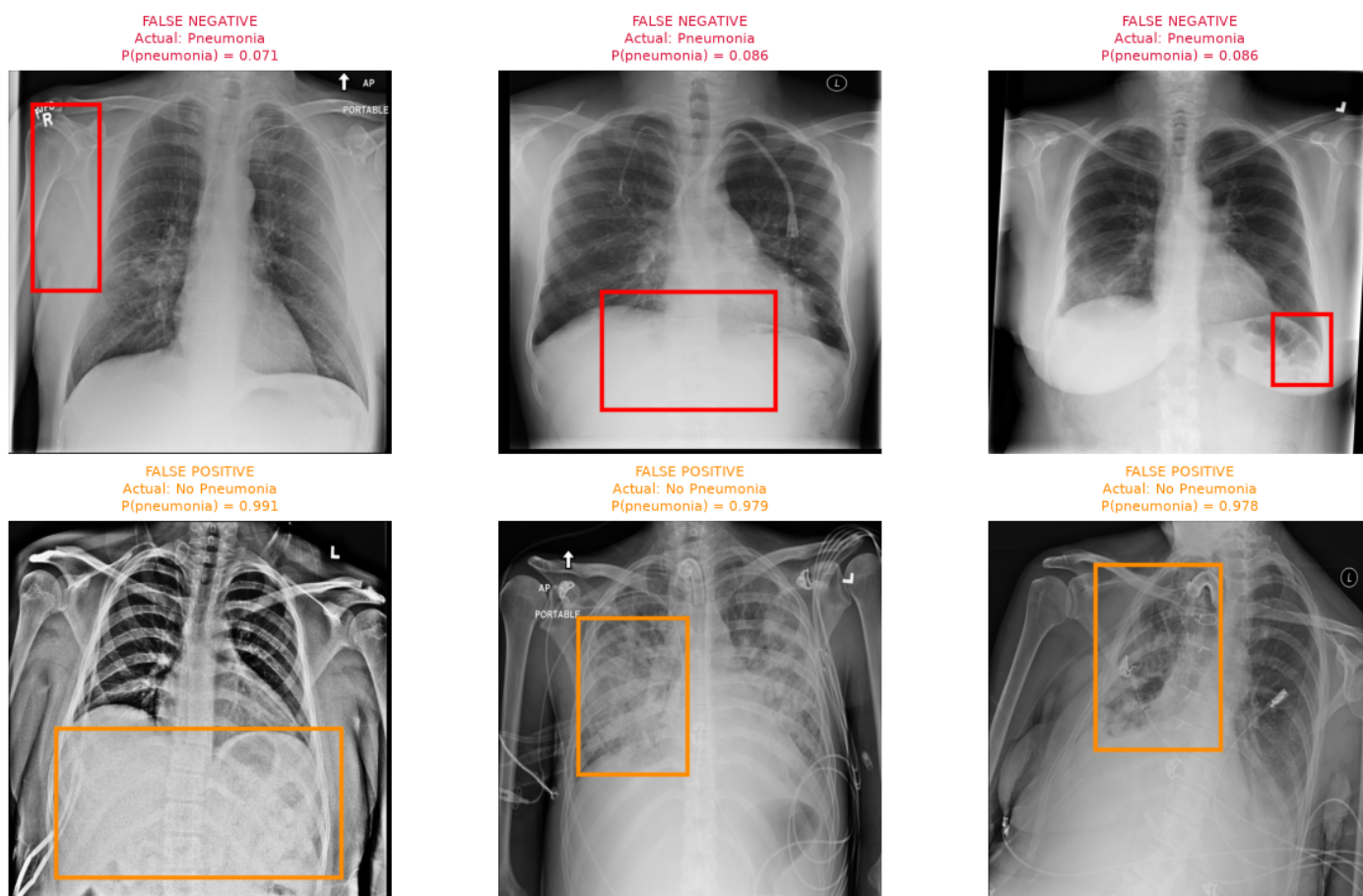


Figure 9 — Misclassified samples. False negatives (top) tend to show subtle or early-stage opacities. False positives (bottom) often come from the 'No Lung Opacity / Not Normal' sub-class.

10. Key Findings & Business Implications

10.1 Technical Findings

- True pneumonia prevalence (22.5%) is lower than the raw CSV implies (31.6%).** The label file contains per-bounding-box rows, not per-patient rows. Deduplication before splitting is essential to avoid biased class ratios.
- ROC-AUC of 0.77 confirms the model learns real signal.** Despite near-zero recall at threshold=0.5, the model correctly ranks many pneumonia cases above non-pneumonia cases — the issue is the threshold, not the representations.
- The 0.5 threshold is inappropriate for imbalanced medical classification.** A lower threshold (0.2–0.3) would recover substantially more recall at the cost of precision — an acceptable trade-off in a screening context where false negatives are far more costly than false positives.
- 128×128 resolution loses critical radiological texture.** Downsampling from 1024×1024 to 128×128 (64× fewer pixels) removes fine-grained opacity patterns. The 224×224 resolution used by standard pretrained backbones will capture significantly more detail.
- 'No Lung Opacity / Not Normal' is the hardest negative sub-class.** At 44.3% of all patients, this class contains real chest abnormalities that can resemble pneumonia visually, making it the primary driver of false positives.

10.2 Business Implications

Finding	Business Impact
1% pneumonia recall at threshold=0.5	Not deployable as-is — would miss 99% of cases, offering no clinical value
ROC-AUC of 0.77	Meaningful baseline — validates the pipeline and benchmarks against transfer learning
Runs on consumer hardware (M1 Mac)	Low infrastructure cost; viable for edge deployment in resource-constrained settings
~2 minutes per training epoch on M1	Full retraining feasible overnight; fast iteration cycle for fine-tuning
Consistent 22.5% stratified splits	Reliable evaluation framework — performance numbers are reproducible and unbiased

Recommended threshold for screening use: ROC curve analysis suggests that a threshold of approximately **0.25–0.35** would balance recall with a manageable false positive rate — a decision that should be made with clinical stakeholders, not purely on statistical grounds.

11. Limitations & Future Work

11.1 Current Limitations

- **128×128 input resolution** discards most radiological detail present in the original 1024×1024 DICOM images.
- **CNN from scratch** (126K parameters) has insufficient capacity to learn complex hierarchical features without pre-trained weights.
- **No threshold calibration** was performed — recall at the default threshold is clinically unacceptable.
- **No data augmentation** was applied during training.

11.2 Planned for Final Report

Task	Approach
Transfer learning (≥2 models)	VGG16, ResNet50, EfficientNetB0 with 224×224 input; freeze backbone → train head → unfreeze last N lay

Task	Approach
Custom architectures	Add attention layers and custom classification heads on pretrained backbones
Threshold calibration	Plot precision-recall curve; select threshold maximising F1 or recall at acceptable precision
Model comparison	Side-by-side AUC, recall, precision, F1 table for all models
Best model serialization	Save in .keras format; reload and run inference on sample images
Web application	Streamlit/Gradio app: upload DICOM or PNG → predicted class + probability
Containerisation	Docker image with model + app; docker run -p 8501:8501
Deployment	Hugging Face Spaces (Gradio)

12. Conclusion

This interim report establishes a complete, reproducible deep learning pipeline for automated pneumonia detection from chest X-ray images:

- **Data ingestion** from DICOM zip archives with handling of dataset-specific quirks (missing DICM headers, macOS ghost files, per-bounding-box duplicate rows).
- **EDA** revealing the true class distribution (22.5% pneumonia at patient level, not 31.6% as the raw CSV implies) and the visual characteristics that distinguish each class.
- **Preprocessing** producing stratified 70/15/15 patient-level splits with verified [0,1] normalization.
- **Baseline CNN** achieving ROC-AUC 0.7676 — a meaningful signal above random, confirming that deep learning can discriminate pneumonia from non-pneumonia in this dataset.

The baseline model is a deliberately lightweight starting point. Its limitations — particularly the 1% pneumonia recall at the default threshold — motivate the transfer learning approach in the final report, where pretrained backbones are expected to push AUC above 0.85 and deliver clinically meaningful recall, bringing the system closer to its goal of serving as a reliable automated decision-support tool for pneumonia diagnosis.

Notebook: src/pneumonia_interim.ipynb | Best model checkpoint: src/best_cnn_model.keras